Complex Optical Remote-Sensing Aircraft Detection Dataset and Benchmark

Tianjun Shi, Jinnan Gong, Shikai Jiang, *Member, IEEE*, Xiyang Zhi, Guangzhen Bao, Yu Sun, and Wei Zhang

***Abstract*— Aircraft detection in remote-sensing images is significant in both military and civilian fields, such as air traffic control and battlefield dynamic monitoring. Deep-learning (DL) methods can achieve promising detection performance with sufficient and labeled samples. However, current aircraft datasets are mainly from a single data source and lack diverse scenes and targets, making it difficult to train a robust and generalized detector. Therefore, we manually label and construct a complex optical remote-sensing aircraft target detection dataset from Google Earth and multiple satellites such as WorldView-2, WorldView-3, Pleiades, Jilin-1, and IKONOS. It contains 7337 images covering typical airports and various rare scenes, including the aircraft carrier, ocean, and land with flying aircraft. The dataset consists of 32 285 civil and military aircraft instances, including bombers, fighters, and early warning aircraft. These targets range from 4 × 4 to 240 × 240 pixels and are all labeled with both horizontal bounding box (HBB) and oriented bounding box (OBB) annotations. The various scenes and sufficient instances can fully support the training and evaluation of data-driven algorithms. Meanwhile, based on the constructed dataset, we train and evaluate several detectors to provide a benchmark and help promote the development of aircraft detection techniques.**

***Index Terms*— Aircraft detection, benchmark, dataset, optical remote sensing.**

I. Introduction

RECENTLY, countries worldwide begin to focus more and more on airspace control, thus promoting the detection of targets with a high strategic value represented by aircraft as a research hotspot. The development of aircraft detection has considerable significance in civilian and military applications [1]. In the civilian field, aircraft detection provides technical support for air traffic control, which helps with flight arrangement and route planning [2], [3]. In the military field, accurate detection of fighters, bombers, and other aircraft helps to analyze the military dynamics. It is beneficial for early warning, military deployment, and battlefield defense [4], [5].

The development of deep-learning (DL) techniques has dramatically improved aircraft detection performance in remote-sensing images. Through trainable feature extractors such as convolutional neural networks (CNNs) [6] and transformers [7], these methods can effectively extract robust features by mining helpful information from training samples adaptively. With a well-designed network architecture, DL-based methods can achieve promising detection performance with high accuracy. However, these methods rely on sufficient and labeled samples [8]. Current aircraft datasets are homogeneous in terms of data sources, imaging scenes, and target types, which makes it challenging to perform well in practical applications. The available aircraft datasets suffer from the following problems.

1) Data sources of existing aircraft datasets are limited to a single imaging platform, most of which are generated from Google Earth or a single satellite payload. Due to the complexity of remote-sensing imaging chains, the differences in optical and electronic systems can lead to distinctive imaging characteristics in terms of aberration, grayscale, and noise distribution [9], [10], [11]. Meanwhile, the occlusion and blurring degree vary from the acquisition angle, weather conditions, and other factors. Therefore, it is difficult to transfer the detection model trained from one domain to the others.
2) The imaging scenes and target types are relatively scarce. The existing aircraft datasets are mainly composed of the apron and runway scenes in civil airports. Targets in these scenes with little interference from false alarm sources are easy to detect. At the same time, the target types are mainly civil aircraft lacking military aircraft samples, which makes the feature characterization simple. The shortage of complex scenes and multiple target samples restricts the practical application of aircraft detection methods.

The above problems are not conducive to improving the robustness and generalization of the detection models, which affects the development of aircraft detection techniques. Therefore, we construct a brand new complex optical scene remote-sensing aircraft detection dataset (CORS-ADD). The dataset is derived from Google Earth, and several satellites with varying imaging characteristics, such as WorldView-2, WorldView-3, Pleiades, Jilin-1A, Jilin-1B, and IKONOS. Besides the typical airport scenes, the dataset consists of various rare samples such as the aircraft carrier, ocean, and land with flying aircraft. Meanwhile, the types are abundant with multiple targets such as civil aircraft, bombers, fighters, and early warning aircraft. The target scale distribution ranges

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The authors are with the Research Center for Space Optical Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: shitianjun@stu.hit.edn.cn; gongjinnan@hit.edu.cn; jiangshikai@hit.edu.cn; zhixiyang@hit.edu.cn; guangzhenbao@stu.hit.edu.cn; ysun@stu.hit.edu.cn; wzhang@hit.edu.cn).

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TABLE I
SUMMARY OF THE REMOTE-SENSING DATASETS WITH LABELED AIRCRAFT

Dataset	Year	Institution	Data Source	Annotatioon	Instances
NWPU VHR-10	2014	Northwestern Polytechnical University	Google Earth, Vaihingen Dataset	HBB	707
UCAS-AOD	2015	University of Chinese Academy of Sciences	Google Earth	HBB	7482
RSOD-Dataset	2017	Wuhan University	Google Earth, Tianditu	HBB	4993
DOTA	2017	Wuhan University	Google Earth, Jilin-1, GaoFen-2	OBB	14085
LEVIR	2018	Beijing University of Aeronautics and Astronautics	Google Earth	HBB	4724
DIOR	2018	Northwestern Polytechnical University	Google Earth	HBB	10104
RarePlanes	2020	IQT-CosmiQ Works	WorldView-3	HBB	14700
FAIR1M	2021	Aerospace Information Research Institute	Google Earth, GaoFen	OBB	30324
HRPlanes	2022	Yildiz Technical University	Google Earth	HBB	18477
CORS-ADD (ours)	2023	Harbin Institute of Technology	Google Earth, WorldView-2, WorldView-3, Jilin-1A, Jilin-1B, Pleiades, IKONOS	HBB, OBB	32285

from 4 × 4 to 240 × 240 pixels, covering the small, medium, and large scales. All the instances are labeled with both horizontal bounding box (HBB) and oriented bounding box (OBB) annotations. Based on the constructed dataset, we provide a benchmark with several detectors to build a research basis for aircraft detection.

The main contributions of our work can be summarized as follows.

1) A complex optical remote-sensing aircraft dataset is constructed, which originates from several satellites with varying imaging characteristics and covers more diverse aircraft samples, thus supporting the training and evaluation of data-driven algorithms.
2) An aircraft detection benchmark is provided based on the constructed dataset with the typical detection methods trained and evaluated, which can help the advancement and application of aircraft detection techniques.

The remainder of this article is organized as follows. Section II introduces the related works of aircraft datasets and detection methods. Then, we elaborate on the construction process and properties of the proposed aircraft dataset in Section III. Section IV provides a benchmark of aircraft detection with several typical detectors. Finally, we summarize and conclude in Section V.

II. RELATED WORKS

A. Aircraft Dataset

Through mining the practical information of data, DL-based methods can learn to extract robust and effective features of the target different from the background. However, such methods are difficult to achieve promising performance in the absence of samples. With the development of remote-sensing imaging technology, more samples are prone to be acquired and labeled, which promotes the advancement of detection algorithms.

NWPU VHR-10 [12], constructed by Northwestern Polytechnical University, originates from the Google Earth and Vaihingen dataset, of which 88 images contain 757 aircraft instances. University of Chinese Academy of Sciences provides the UCAS-AOD [13], which is derived from Google Earth containing 7482 instances. Wuhan University introduces the RSOD-Dataset [14] with four classes, including aircraft, playground, overpass, and oil barrel. This number of images containing aircraft is 446 with 4993 instances. DOTA [15] is generated from Google Earth, JiLin-1, and GaoFen-2, which contains 15 classes with 14 700 aircraft instances. LEVIR [16] originates from Google Earth and is provided by the Beijing University of Aeronautics and Astronautics with 4724 aircraft instances. DIOR [17] contains 20 types of targets, including 10 104 aircraft instances. RarePlanes [18] is constructed by IQT-CosmiQ Works from Maxar WorldView-3 and synthetic data, with a total of 14 700 manually annotated instances. Constructed by Aerospace Information Research Institute, FAIR1M [19] from Google Earth and Gaofen satellite with 30 324 civil aircraft instances. HRPlanes [20] is a recently released aircraft detection dataset whose data source is Google Earth with 18 477 instances. Table I illustrates the statistical results of our research on aircraft datasets. It is obvious that current datasets are often derived from a single data source, mainly from Google Earth. The homogeneous characteristics of targets and scenes make it difficult to ensure the generalization and robustness of networks. Therefore, we construct a complex optical scene remote-sensing aircraft detection dataset from Google Earth and multiple satellites such as WorldView-2, WorldView-3, Jilin-1, Pleiades, and IKONOS to ensure the diversity of samples.

B. Aircraft Detection Method

The early aircraft detection methods are based on artificially designed feature extractors, which focus on shallow features, including geometric contour, texture, and symmetry [21]. However, such hand-crafted features are easily affected by imaging factors such as lighting and scale. These methods have limited representation ability of target characteristics and can only be applied in specific scenes. DL-based methods can learn to extract more robust features through adaptively mining the effective information of samples. With the proliferation of hardware computational capabilities and data, these methods start to be widely used in remote-sensing target detection [22].

According to whether generating the anchors, DL-based detection methods can be categorized into anchor-based and -free methods [23]. Anchor-based methods require fine-tuned anchor parameters depending on the aspect ratio of targets in the dataset to achieve promising performance. Furthermore, they can be classified into one- and two-stage methods according to whether generating candidate regions. Ding et al. [24] combine online hard example mining (OHEM) and multiscale feature representation with Faster R-CNN to improve the detection performance of targets in remote-sensing images. Wu et al. [25] introduce improved self-calibrated convolution (SC-Conv) and dilated convolution to Mask R-CNN, achieving better performance for aircraft detection. Zeng et al. [26] propose a top-down aircraft detection method in large-scale

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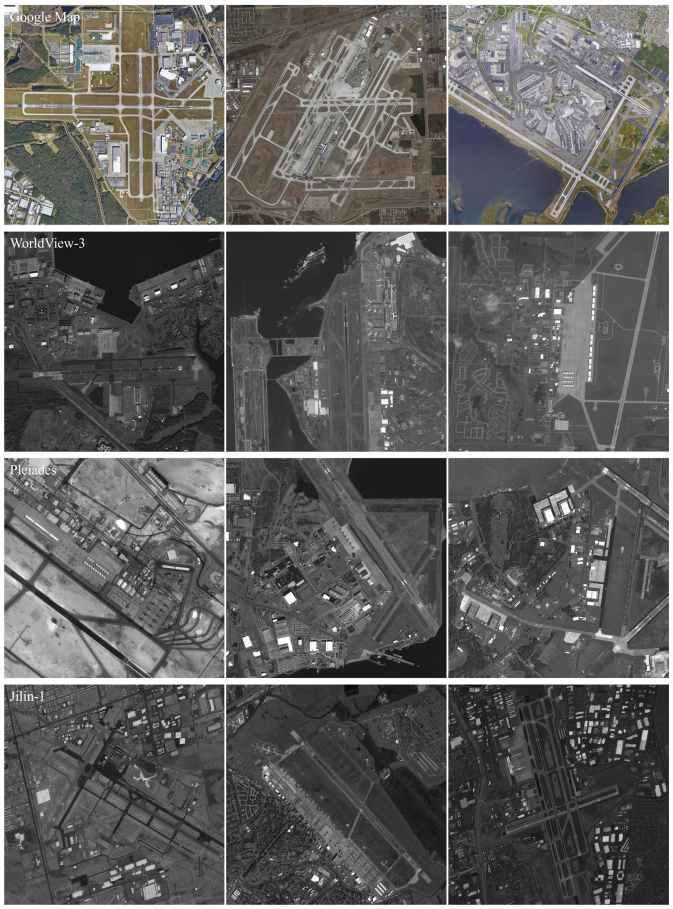


Fig. 1. Display of the airports from different data sources in CORS-AOD.

images, with U-Net for airport search and a feature enhancement Faster R-CNN (FEF-R-CNN) for target detection. Based on the framework of YOLO, Xu and Wu [27] utilize an improved DenseNet to avoid gradient fading and increase the prediction layer from three to four, thus putting forward MRFF-YOLO. Yang et al. [28] introduce generalized intersection over union (GIoU) loss in the bounding box function to YOLOv3 to improve the regression accuracy for aircraft detection. Liu et al. [29] propose you can look more (YOLM) combined with an enhanced neck network to extract more detailed information and attention mechanism to reduce the interference of environmental factors. Anchor-free methods avoid setting hyperparameters about anchors and apply key points to generate the candidates. X-LineNet [30] converts aircraft detection into predicting and grouping of paired intersecting line segments, allowing the network to learn richer and specific visual semantic information. CenterFPANet [31] introduces the feature pyramid and attention (FPA) mechanism to alleviate the difficulty in target detection for large-scale remote-sensing images. Scene-relevant anchor-free networks (SRAF-Net) [32] improves the detection accuracy by combining scene-enhanced feature pyramid network (FPN) with scene refine module to capture the semantic information. The problem lies in that a complete dataset is required for the DL-based methods to achieve promising performance. Due to the homogenization of existing aircraft datasets in the data source, networks are hard to train adequately and evaluate comprehensively. Therefore, based on the constructed dataset, we evaluate multiple algorithms and provide a benchmark to promote the development of aircraft detection methods.

TABLE II
SUMMARY OF THE DATA SOURCE

Data Source	Resolution (m)	Average Size (pixel)	Number of the Large-scale Images
WorldView-2	0.50-0.51	(7680, 7680)	4
WorldView-3	0.31-0.41	(8704, 9842)	9
Pleiades	0.50	(5120, 5120)	5
Jilin-1A	0.72	(4835, 5178)	15
Jilin-1B	0.92	(3072, 4096)	5
IKONOS	1.00	(11721, 12484)	1
Google Earth	0.54-1.02	(8964, 7240)	119

III. Dataset Construction

A. Construction Method

To ensure the diversity of samples, we choose Google Earth and several satellites as the data source, including WorldView-2, WorldView-3, Pleiades, Jilin-1A, Jilin-1B, and IKONOS. Table II illustrates the main parameters of the acquired data, and the resolutions range from 0.31 to 1.02 m. Due to the differences in optical system aberration, detector noise, platform vibration, and other factors in the imaging chains, the targets and scenes exhibit distinctive characteristics. The selected scenarios cover typical civil airports in China and the United States, such as Beijing Capital International Airport, Charlotte Douglas International Airport, Harbin Taiping International Airport, and so on. At the same time, we choose some important air force bases to obtain typical military aircraft, such as Whiteman Air Force Base, Naval Station Norfolk, Kadena Air Force Base, and so on. The images from different satellites are shown in Fig. 1. Actually, except that samples from Google Earth are RGB images, those from satellites we choose to construct the dataset are grayscale images. The proportion of RGB and grayscale images is around 3:1.

The process of constructing the dataset is shown in Fig. 2. First, all the large-scale remote-sensing images are manually labeled by LabelImg2 [33]. We equip all the instances with HBB and OBB annotations to make our dataset easily applied by the mainstream detection networks. To make our dataset more convenient to be used, we offer two kinds of HBB annotation formats: YOLO format and COCO format. The former one denotes each target as (x_c, y_c, w, h), where x_c,y_c,w,h represents the normalized central coordinates, width, and height of the target. The latter one provides the top-left corner and bottom-right corner coordinates of the target, which is denoted as (x_1, y_1, x_2, y_2). For OBB annotations, we provide the standard DOTA format with denotation as $(x_{o1}, y_{o1}, x_{o2}, y_{o2}, y_{o3}, y_{o3}, y_{o4}, y_{o4})$, which represents the four coordinates of the OBB.

Then we cut the images into small blocks and only retain those with targets. Actually, when cropping large-scale images, it is likely to split the targets into two pieces when located at the boundary. The direct cropping operation makes the dataset

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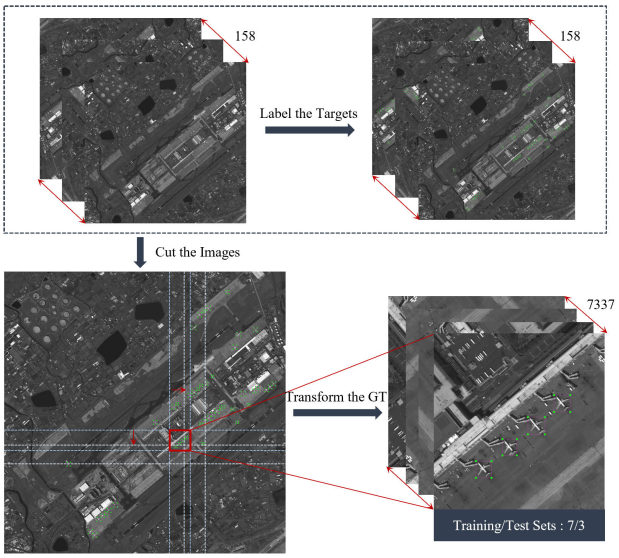


Fig. 2. Process of constructing the dataset.

lose plenty of samples with full geometric shapes. Therefore, we set the 0.25 overlap ratio of each block to ensure each target can maintain at least a complete instance in the dataset. It certainly helps to improve the diversity of the samples. Finally, we transform the ground truth to generate the labels. In fact, this process is to convert the coordinates of targets from large-scale images into small blocks. The conversion relationship is formulated as follows:

$$(x_{c2}, y_{c2}) = (x_{c1}, y_{c1}) - (x_{\text{block}}, y_{\text{block}}) \tag{1}$$

where (x_{c2}, y_{c2}) and (x_{c1}, y_{c1}) represent the central coordinates of the target in the large-scale images and blocks, respectively. $(x_{\text{block}}, y_{\text{block}})$ is the top-left corner coordinate of the block in the large-scale images. Also, we set a threshold to decide whether to reserve the targets located at the boundary. When the remaining part of a target is less than this threshold, we consider this area lacking abundant information to characterize the target and abandon the label.

To sum up, 158 large-scale images with an average size of 8234 × 7070 pixels are labeled, which can be used for algorithm deployment in practical applications. The number of image blocks is 7337, with a size of 640 × 640 pixels and a total of 32 285 aircraft instances. These images are divided into the training set and test set with a ratio of 7:3, which can be used to train and evaluate the detectors. Fig. 3 displays the annotations of Davis-Monthan Air Force Base and Pearl Harbor.

B. Dataset Properties

As shown in Fig. 4, the constructed dataset contains various complex scenes in which aircraft may appear. Fig. 4(a)–(d) displays the samples in the apron and runway. Obviously, samples derived from different satellites present great differences in the resolution, grayscale, texture, and other features of the targets and scenes. Despite typical airport scenes, the dataset also covers several rare scenes, such as the aircraft carrier with fighters and early warning aircraft in Fig. 4(e). At the same time, samples under different weather and imaging conditions are included in the dataset. Fig. 4(f) and (g) shows the samples under severe degradation of the imaging chains with blurred texture and weakened edge features. Fig. 4(h) and (i) and (j) and (k) displays the samples with cloud interference and low dynamic range, respectively. The above examples increase the difficulty of aircraft detection and help to improve the generalization and robustness of the detectors. In addition to parked aircraft, several moving aircraft are included in this dataset. For example, Fig. 4(l)–(o) shows the aircraft flying on the sea and land.

Apart from the common civil aircrafts such as B737 and B747, our constructed dataset also covers various military targets such as bombers, fighters, and early warning aircraft to meet the requirement of different applications. As shown in Fig. 5, the different types of targets originate from multiple data sources to ensure the diversity of target categories. According to the definition in COCO [34], targets occupy smaller than 32 × 32 pixels, from 32 × 32 to 96 × 96 pixels, and larger than 96 × 96 pixels are defined as small, medium, and large targets, respectively. The size distribution histogram of the targets in the dataset is illustrated in Fig. 6. The dataset contains aircraft ranging from 4 × 4 to 240 × 240 pixels, which is from small to large to ensure the diversity of target scales.

IV. Aircraft Detection Benchmark

A. Evaluation Metrics

To better put the constructed dataset into practical application, we train and evaluate several typical detectors. We provide a benchmark and hope to contribute to the development of aircraft detection methods. Precision, Recall, F1-Score, and average precision (AP) are selected as the evaluation metrics.

1) Precision and Recall: Precision is defined as the percentage of real targets in the predictions from the algorithm, which is used to assess the degree of false alarms. Recall measures the degree of missed alarms, which is the percentage of correct predictions in the labeled targets. They are formulated as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

where TP, FP, and FN refer to true positive, false positive, and false negative, respectively.

2) F1-Score and AP: These metrics can evaluate the detection performance more comprehensively. F1 score is defined as the harmonic mean of the Precision and Recall. AP is calculated as the area enclosed by the curve of Precision and Recall, and the IoU threshold is set to 0.5

$$\text{F1} = \frac{2}{\frac{1}{P} + \frac{1}{R}} \tag{4}$$

$$\text{AP} = \int_0^1 P(R) d(R). \tag{5}$$

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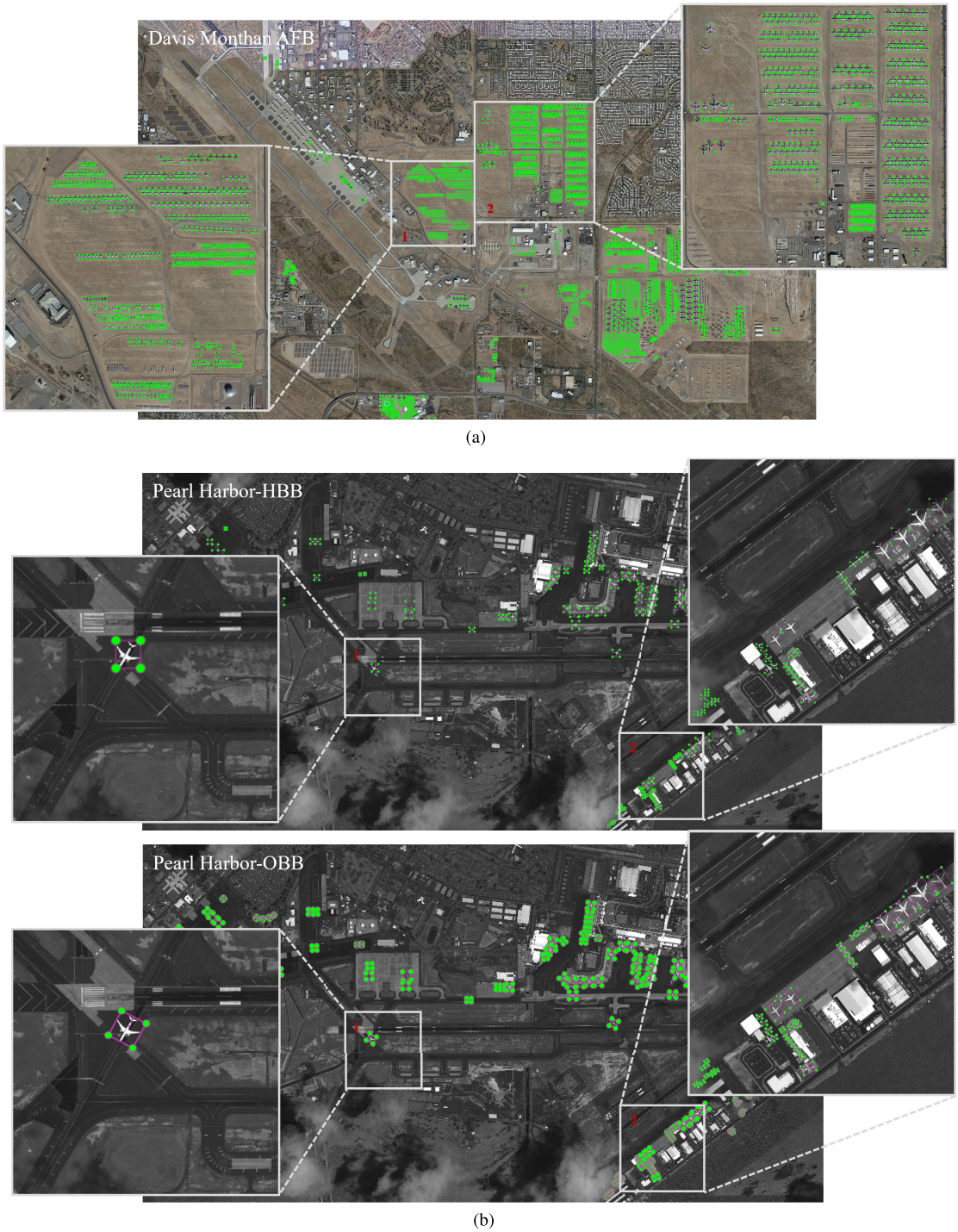


Fig. 3. Display of the labeled large-scale images. (a) Davis-Monthan Air Force Base. (b) Pearl Harbor with HBB and OBB annotations.

B. Results Analysis

DL-based detectors can be classified into anchor-based and -free detectors according to whether requiring predefined anchor parameters or not. Furthermore, anchor-based detectors can be categorized into two- and one-stage detectors. To fully evaluate the performance of existing methods for this dataset, we select three typical detectors from each of the three classes: Faster R-CNN [35], Cascade R-CNN [36], RoI Transformer [37], RetinaNet [38], YOLOv5 [39], S2A-Net [40], FCOS [41], YOLOX [42], and Oriented RepPoints [43]. Faster R-CNN, the most classical two-stage detector, introduces the region proposal network (RPN) to achieve end-to-end object detection and further simplifies the detection pipeline. Cascade R-CNN cascades multiple networks through different IoU thresholds to optimize the prediction results. ROI Transformer introduces a fully connected layer with a dimension of 5 to convert horizontal RoI to oriented RoI. RetinaNet utilizes the FPN and focal loss to improve the detection performance of one-stage detectors. YOLOv5 combines various advanced bags of freebies (BoF) and bags of specials (BoS) to achieve

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TABLE III
EVALUATION RESULTS OF THE TYPICAL DETECTORS

Methods		Annotation	P	R	F1	AP
Two-stage detectors	Faster R-CNN	HBB	94.96%	88.05%	0.9138	0.9312
	Cascade R-CNN	HBB	95.05%	88.51%	0.9166	0.9338
	RoI Transformer	OBB	96.91%	92.38%	0.9459	0.9052
One-stage detectors	RetinaNet	HBB	93.35%	84.95%	0.8895	0.8921
	YOLOv5	HBB	96.23%	92.58%	0.9437	0.9533
	S2A-Net	OBB	96.56%	87.52%	0.9183	0.8999
Anchor-free detectors	FCOS	HBB	94.56%	87.06%	0.9066	0.9017
	YOLOX	HBB	94.82%	90.05%	0.9237	0.9404
	Oriented RepPoints	OBB	94.65%	88.56%	0.9150	0.8996

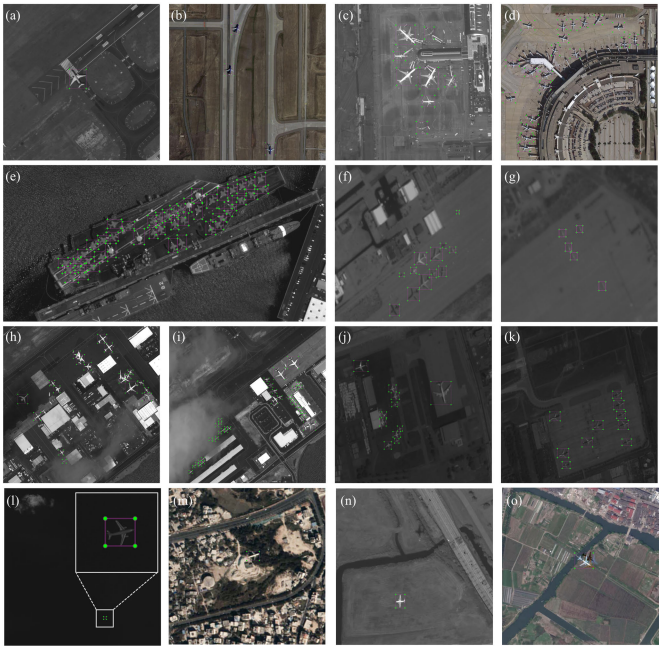


Fig. 4. Display of the complex scenes in the dataset. (a)–(d) In the apron and runway. (e) In the aircraft carrier. (f)–(j) Under severe degradation. (h)–(i) Cloud interference. (j)–(k) Low dynamic range. (l)–(o) Flying aircraft.

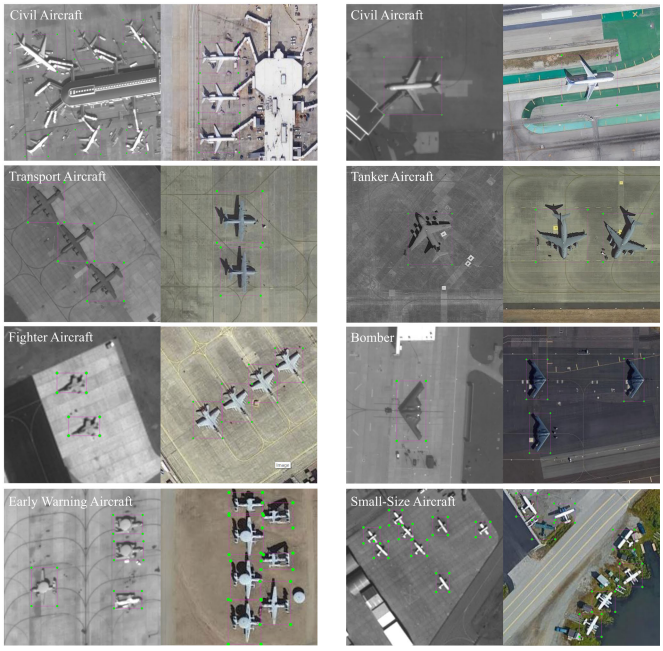
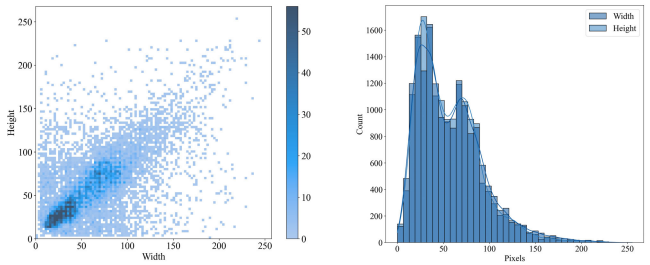


Fig. 5. Display of the aircraft in the dataset.



(a) (b)

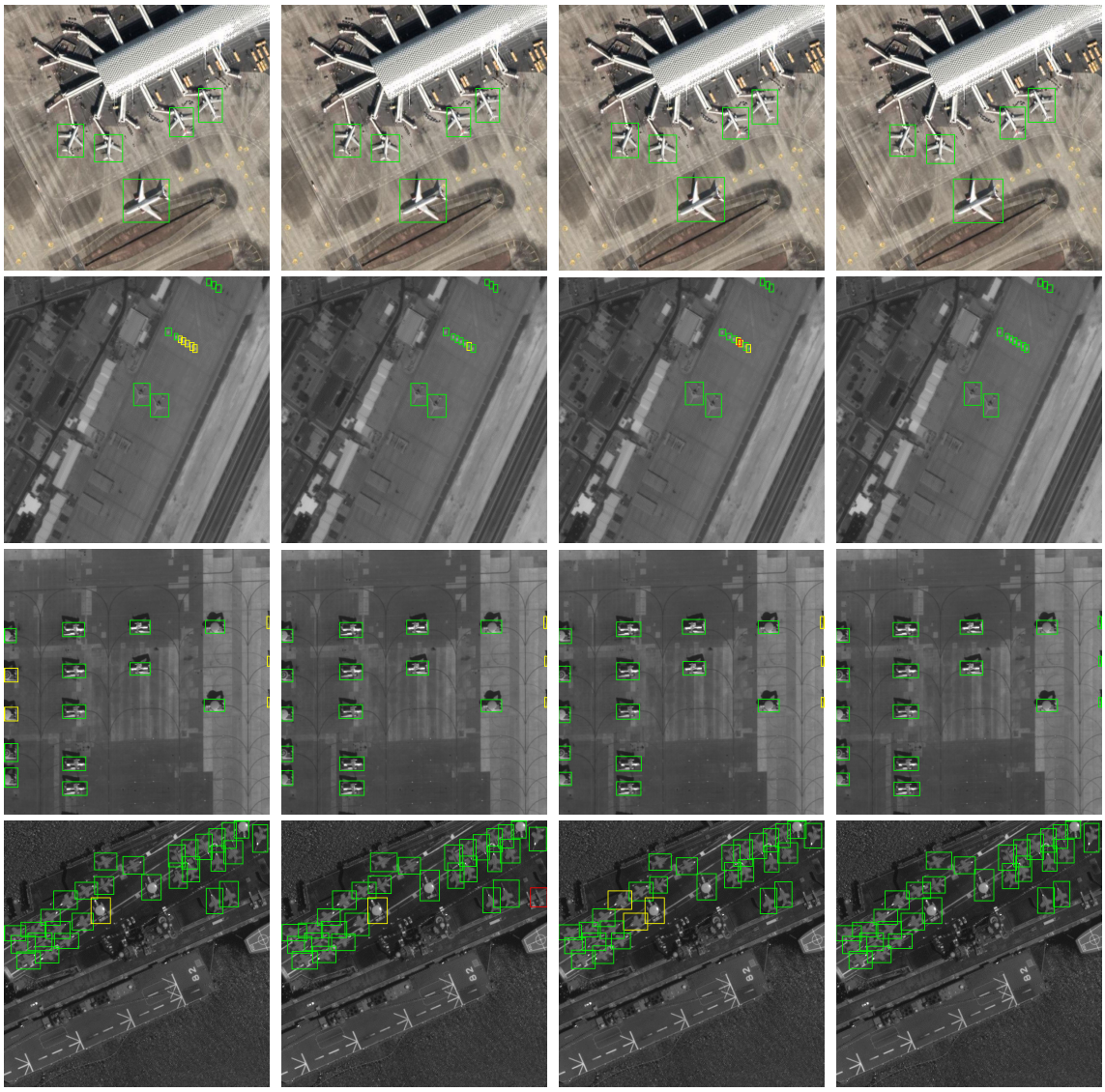
Fig. 6. Scale distribution of the targets in the dataset. (a) Height and width distribution. (b) Area distribution.

a balance between accuracy and speed. S2ANet applies the feature alignment module (FAM) to generate high-quality anchors and the oriented detection module (ODM) to alleviate the inconsistency between the classification score and localization accuracy. FCOS directly predicts the distance from the four sides of the target to the center and utilizes the center-ness prediction branch to suppress the low-quality predicted bounding boxes. YOLOX introduces the anchor-free mechanism to YOLO-based architecture. It adopts decoupled head, SimOTA, and other advanced detection technology tricks to improve performance. Oriented RepPoints introduces the adaptive points representation for diverse orientations, shapes, and poses to fully capture the geometric information. Through the Pytorch framework, all the experiments are carried out on the workstation with an Inter Xeon[1] Silver 4210R CPU and an NVIDIA GeForce RTX 2080 SUPER GPU. We train and evaluate the above nine detectors based on the constructed dataset. The horizontal and oriented detectors are evaluated by YOLOv5 and MMRotate [44] codes, respectively.

[1]Registered trademark.

As shown in Table III, the overall Precision of the two-stage methods is higher than that of the others. The coarse-to-fine pipeline helps to verify the targets and reduce false alarms. With the help of several advanced detection tricks, YOLOv5 and YOLOX outperform the chosen two-stage detectors. Among all the algorithms, YOLOv5 achieves the optimal performance with an AP of 0.9533, which is 2.06% and 1.40% higher than that of Cascade R-CNN and YOLOX, respectively. Obviously, the overall Recall values are relatively low compared to the Precision values, which means that there

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(a) (b) (c) (d)

Fig. 7. Visualization of the detection results, where TP, FN, and FP are shown by the green, yellow, and red rectangles, respectively. (a) Cascade-R-CNN. (b) YOLOv5. (c) YOLOX. (d) Ground truth.

is still great potential for aircraft detection. Fig. 7 displays the visualized detection results of the two-stage Cascade R-CNN, one-stage YOLOv5, and anchor-free YOLOX. All three methods can achieve promising detection performance for civil aircraft in the common airport (the first row of Fig. 7). The confidence of the Cascade R-CNN is higher than the other methods, which proves the effects of the coarse-to-fine framework. YOLOv5 can guarantee excellent performance for small targets, as shown in the second row of Fig. 7. The Recall of Cascade R-CNN is relatively low, so several missed alarms exist in the second and third rows of Fig. 7. For the complex aircraft carrier scene (the fourth row of Fig. 7), all the methods have some degree of missed and false alarms.

C. Discussion

We take YOLOv5 as an example to analyze the current difficulties of aircraft target detection. Due to the complexity of remote-sensing images, false alarms with similar characteristics to targets are prone to appear. They exhibit almost the same geometric shape and grayscale distribution. As shown in Fig. 8(a) and (b), the small fork marks are misjudged as aircraft because of the similar symmetry feature. Meanwhile, the degradation of the imaging link blurs the differential characteristics of targets and scenes, thus making the network difficult to distinguish them. Fig. 8(c) and (d) shows that small artificial facilities are detected as aircraft due to their blurred and confusing characteristics. Besides, small-scale targets with small geometric sizes and weak features are hard to detect, according to Fig. 8(e) and (f). Interfered by complex weather factors such as clouds and fog, small-scale targets are more likely to be ignored by the network, as shown in Fig. 8(f).

To solve the above problems, aircraft detection networks can be improved in the following ways. Multiscale and multilevel feature fusion methods can be improved to integrate the high-level semantic features and low-level detailed features fully. The utilization of contextual information helps to eliminate the interference of complex scenes. At the same time,

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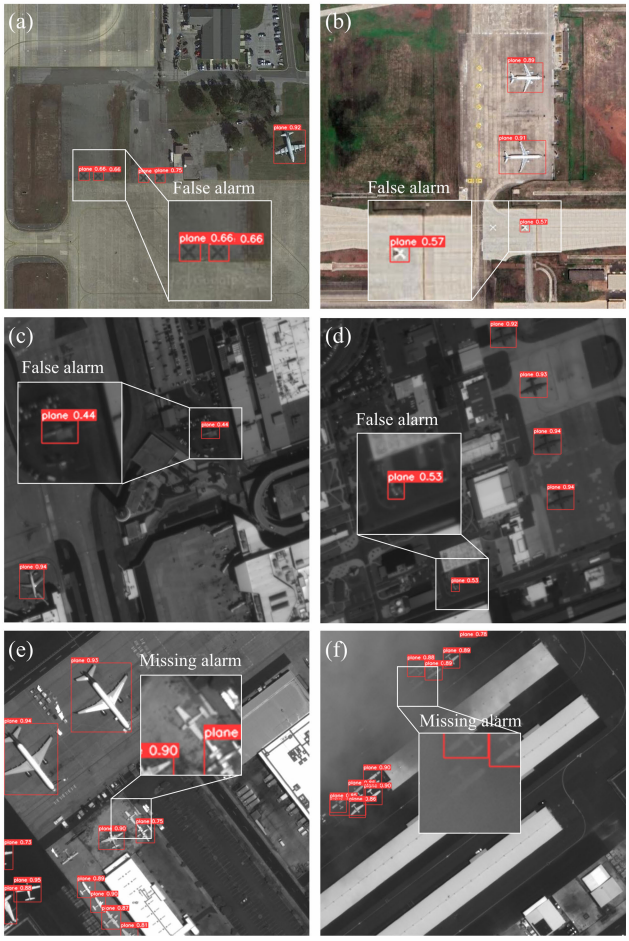


Fig. 8. False and missing alarms of the YOLOv5 detection results. (a)–(d) False alarms. (e)–(f) Missing alarms.

the tailored attention mechanism can be applied to enhance the saliency representation of aircraft in the channel and spatial dimension. It may improve the network's perception of small-scale targets to reduce missing alarms.

V. Conclusion

In this article, we construct a CORS-ADD. Google Earth and multiple satellites are selected as the data sources, like WorldView-2, WorldView-3, Pleiades, Jilin-1, and IKONOS, to ensure the diversity of scenes and targets. We obtain 7337 images based on manual HBB and OBB annotation with a total of 32 285 aircraft. Despite the apron and runway, our dataset also contains various rare scenes, including the aircraft carrier, ocean, and land with flying aircraft. The targets cover multiple aircraft such as civil aircraft, bombers, fighters, and early warning aircraft, of which the scale ranges from 4×4 to 240×240 pixels. Based on the constructed dataset, we provide a benchmark by training and evaluating several typical detectors. We hope our work can help promote the improvement and practical application of aircraft detection technology.

Acknowledgment

Complex optical remote sensing aircraft detection dataset (CORS-AOD) is available at https://github.com/sgtojd/Complex-Optical-Remote-Sensing-Aircraft-Detection-Dataset.

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Tianjun Shi received the B.Eng. and M.Eng. degrees in optical engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2020 and 2022, respectively, where he is currently pursuing the Ph.D. degree in optical engineering.

His research interests include remote-sensing image acquisition and processing and small target detection and identification.



Jinnan Gong received the Ph.D. degree in optical engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2016.

He is currently an Assistant Professor with HIT. His research interests include optical image processing, processing algorithm optimization at the engineering level, and performance evaluation.



Shikai Jiang (Member, IEEE) received the M.Eng. and Ph.D. degrees in optical engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2018 and 2022, respectively.

He is currently an Assistant Professor of optical engineering with the School of Astronautics, HIT. He is interested in image processing of space optical remote-sensing satellites, including image inversion restoration, image fusion, super-resolution, and target detection and identification.



Xiyang Zhi received the Ph.D. degree in optical engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2012.

He is currently a Full Professor with HIT. His research interests include remote-sensing image acquisition and processing and optical target detection and identification.



Guangzhen Bao received the B.Eng. and M.Eng. degrees in optical engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2020 and 2022, respectively.

His research interests include remote-sensing image acquisition and processing and optical target detection and identification.



Yu Sun received the B.Sc. degree in optoelectronic information science and engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2020, where he is currently pursuing the Ph.D. degree.

His research interests include image processing of space optical remote-sensing satellites such as super-resolution and restoration.

Wei Zhang received the M.S. degree in optical engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 1986, and the Ph.D. degree in electronic engineering from the Tohoku Institute of Technology, Sendai, Japan, in 2000.

He is currently a Full Professor with HIT. His research interests include remote-sensing image acquisition and processing, optical system design, and automatic target detection and identification.

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